Learning a Policy Repository through Adaptive Environment Assignment

June 2026

EPFL



OVERVIEW

- Motivation
- Related Work
- Methods
- Results
- Limitations and Future Work

MOTIVATION

OPEN-ENDED LEARNING ENVIRONMENTS

Many real-world environments are open-ended:

- Agents continually encounter novel tasks and environments
- Successful behavior requires continual adaptation rather than convergence to a fixed policy



Open-Ended



Fixed-Task

Hughes et al., 2024 PMLR

INTRODUCTION

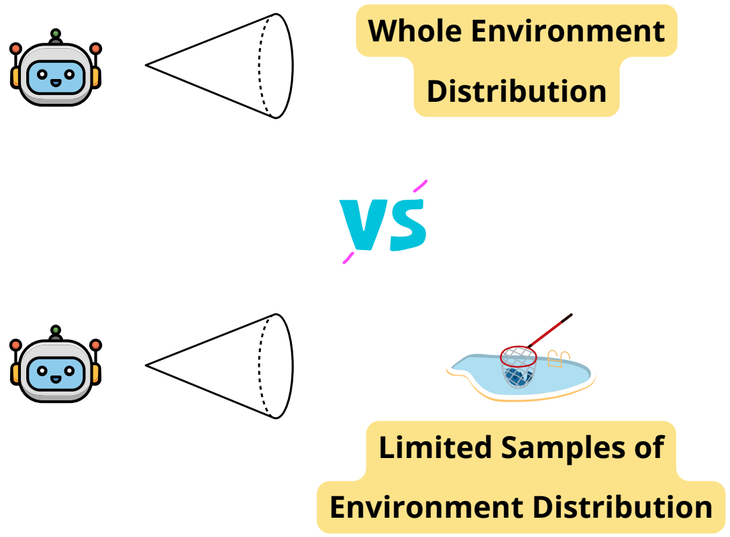
DESIRABLE AGENT PROPERTIES IN OPEN-ENDED LEARNING

A agent in such environments should:

- Avoid overfitting to specific environments
- Transfer previously learned behaviors to novel tasks
- Accumulate knowledge over time
- Preserve prior capabilities without catastrophic forgetting

MOTIVATION

GRADUAL EXPOSURE TO THE ENVIRONMENTS IN OPEN-ENDED LEARNING



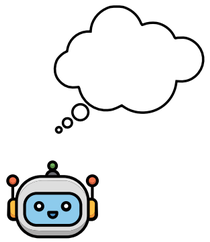
Training on Many Environments in Parallel

\+ Reduces chance of over-specialization / migitates catastrophic forgetting

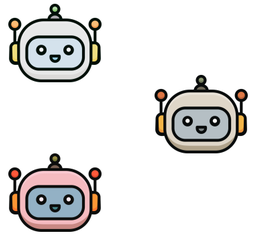
\- Requires access to full environment distribution (unrealistic in open ended learning)

MOTIVATION

AN APPROACH TO INCREASE STABILITY IN OPEN-ENDED LEARNING



Let’s use a prior relevant
to current environment
as training start point!



Specialized Priors

Forming Repository of Policies Each Versed in a Certain Environment Niche

+ Reduces interference that arises when a single policy must represent diverse behaviors.
+ Preserves previously acquired expertise, mitigating catastrophic forgetting.
- Requires identifying envrionment niches

RELATED WORK

Unsupervised Skill Discovery

Goal: Learn diverse behaviors without task-specific rewards

Common Approach: Learn a fixed set of skills using diversity-based objectives

Limitations

- Behavioral diversity does not necessarily imply task usefulness or transferability
- Learning a fixed set of skills at the beginning of training conflicts with open-ended learning objectives

Example of diverse locomotion skills emerged
by encouraging state-transition diversity
Sharma et al., DADS



Eysenbach et al., 2018, ICLR / Sharma et al., 2020 ICLR / Bagaria et al., 2021, MLPR

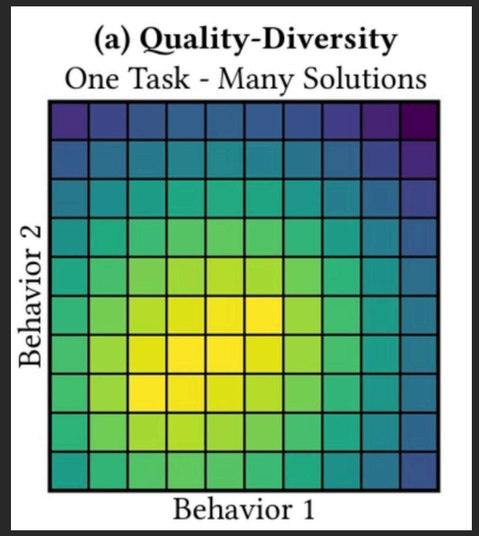
RELATED WORK

Quality Diversity Algorithems

Goal

Discover a diverse set of high-performing solutions rather than a single optimum

Diversity + Quality



Approach

- Split the solution space using behavior descriptions
- Keeping the best-performing policy in each sub-space

Mouret et al., 2015 arXive / Anne et al., 2024 GECCO '24

RELATED WORK

Quality Diversity Algorithems

Challenge:

Diversity emerges w.r.t the chosen behavioral descriptors.

But:

Designing descriptors that correspond to reusable and transferable behaviors is non-trivial.

SCOPE

Evolving Policy Repository Framework

- Learn a diverse repository of priors and match them to novel environments
- Policies evolve over the course of environment encounters rather than being fixed after inital discovery

Gradual Enounter with Environments

Partial access to the environment pool

Out of Scope

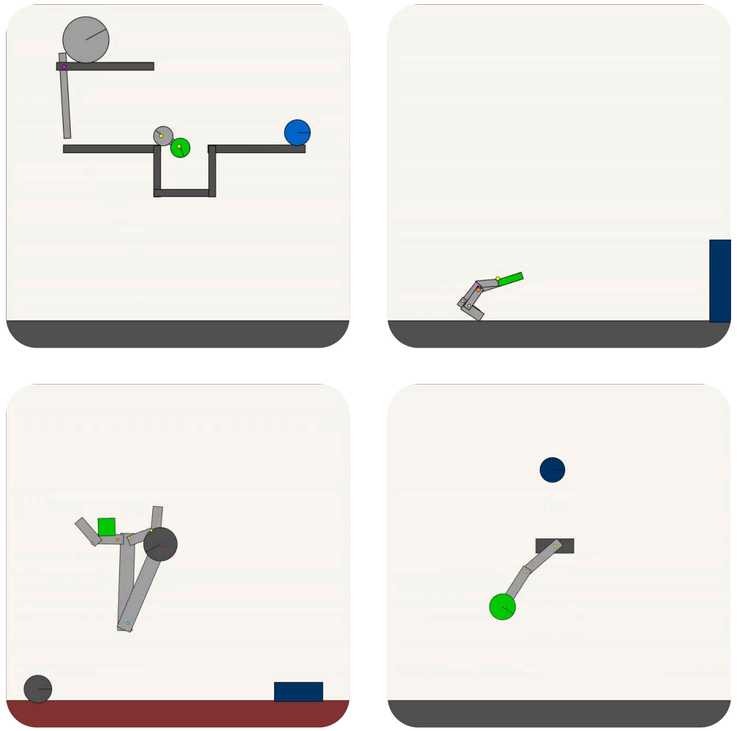
Hierarchical control / Curriculum shaping

METHODS

LEARNING ENVIRONMENT: KINETIX

Attractions:

- Simple goal, diverse tasks
- Possibility to hand-design new environments



Unified Goal

Make the green shape touch the blue shape, without it touching the red shape

🌎 **State space:** full information on available entities in the scene

🤖 **Action space:** controlling on/off and direction for actuators

💎 **Reward:** Getting blue and green closer and colliding them

Matthews et al., 2025 ICLR

METHODS

LEARNING ENVIRONMENT: KINETIX

Policy architecture: Transformer & GNN hybrid → inductive bias on object relationships

Training Algorithm: PPO

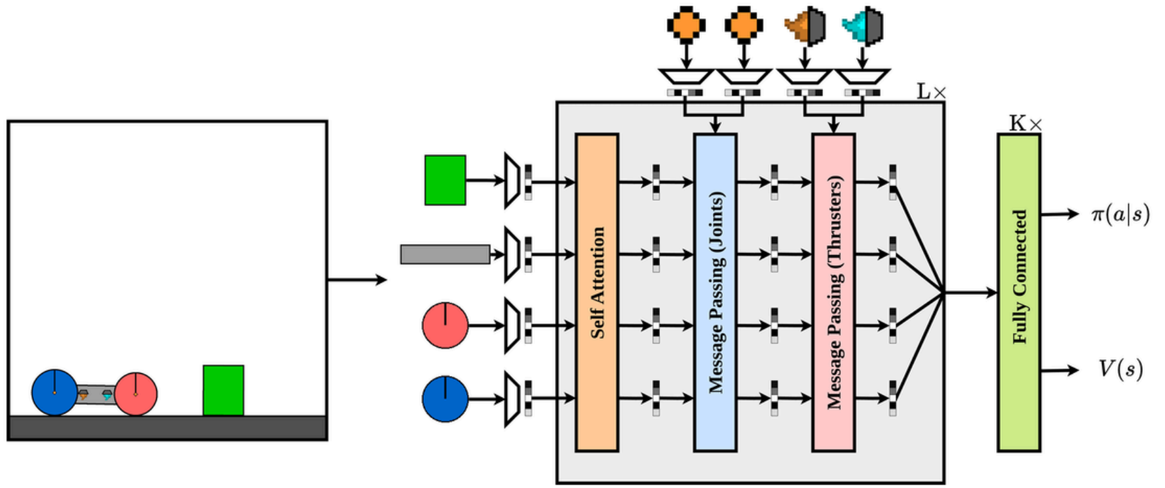


Figure 2: The transformer-based architecture used for training. The scene is decomposed into its constituent entities and then passed through the network, consisting of L layers of self-attention and message passing, followed by K fully connected layers.

Matthews et al., 2025 ICLR

METHODS

LEARNING ENVIRONMENT: TEST SUIT DESIGN

No Explicit Grouping of Environment Space in The Original Test Suite

Test on Systematic Perturbations of The Environments



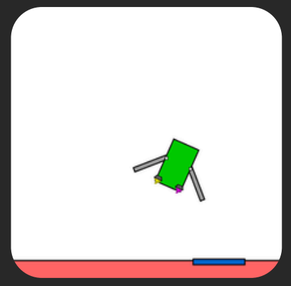
METHODS

FINAL GOAL: A DIVERSE SET OF GOOD STARTING POINTS

Repository of Policies

$$\Pi = \{\pi_1, \pi_2, \ldots, \pi_N\}$$

Environment



Routing

π_s

Selected Policy
(To be finetuned on the env)

Among available policies, which one is more likely to be successful on a given environment?

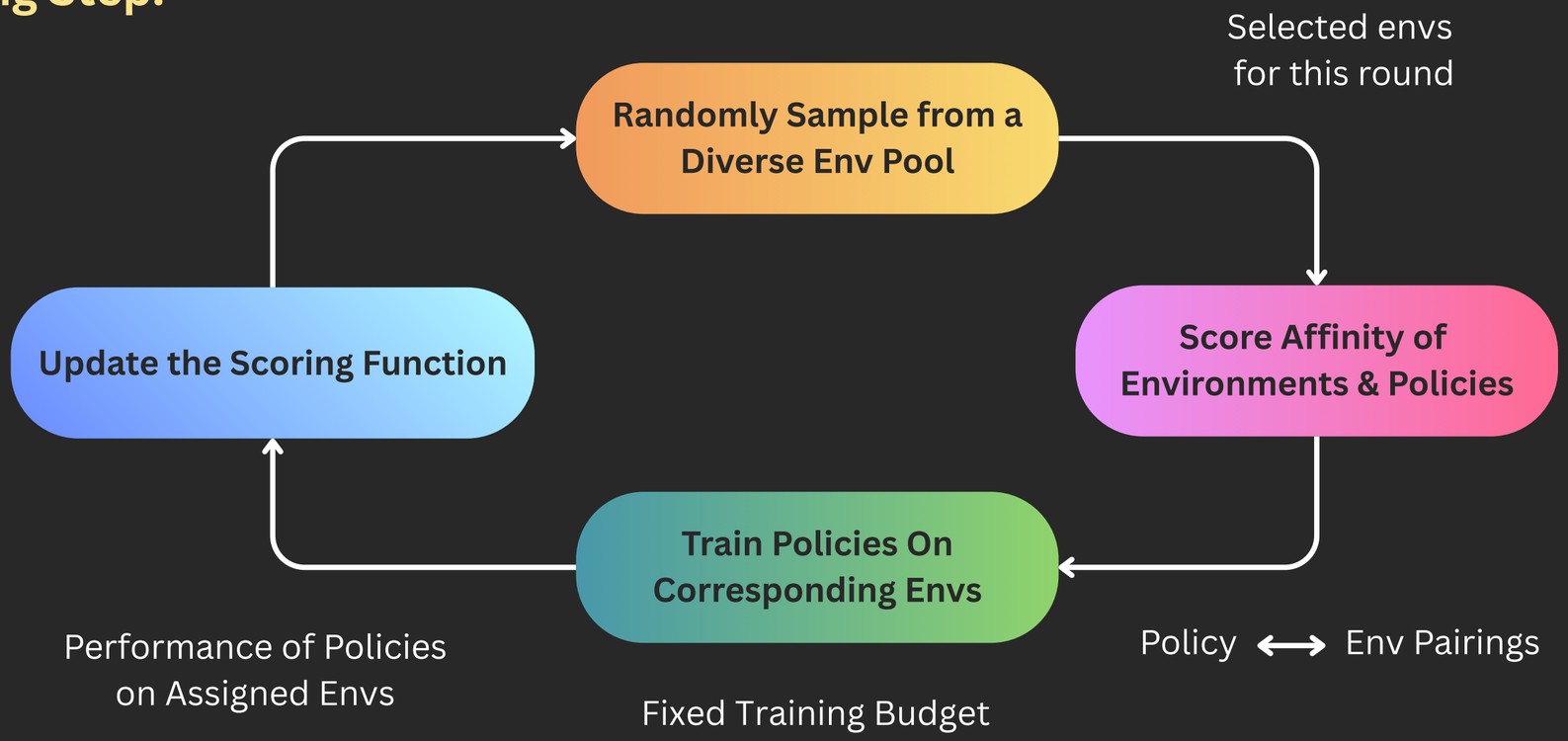
[Analogous to Mixture of Experts]

METHODS

POLICY REPOSITORY: TRAINING LOOP

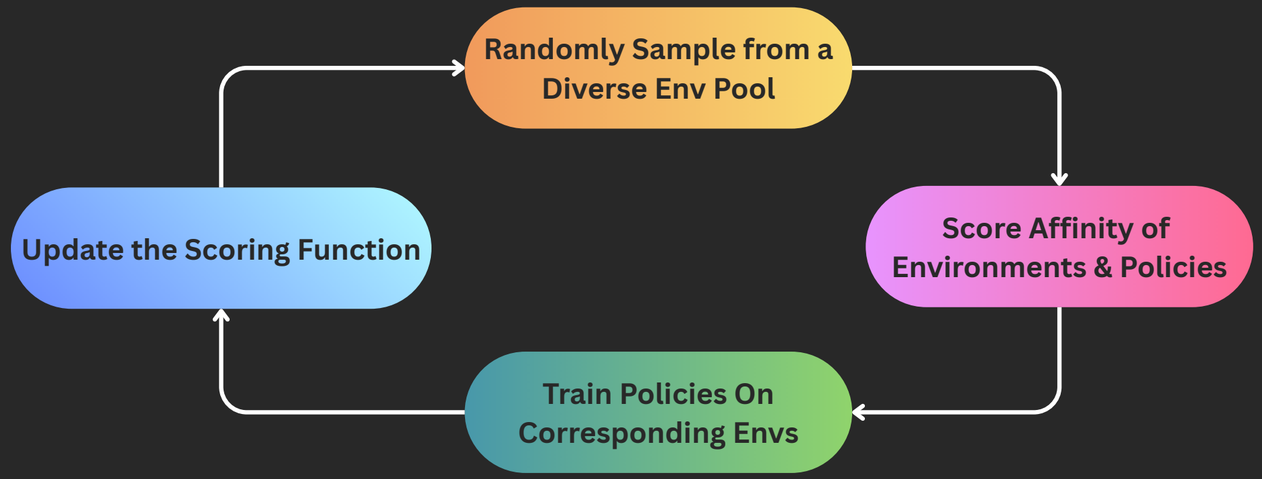
Start with random policies; Get them specialized throughout repository training

One Training Step:



METHODS

POLICY REPOSITORY: TRAINING LOOP



We can play with
Scoring Function!

Role of Scoring Function:
Have the policy getting trained on envrionemnts not far from previous ones, implicitely leading to specicalization in a niche

Specilization Mechanism:
Environment Assigment

METHODS

ENVIRONEMNT ↔ POLICY AFFINITY SCORING

Repository of Policies

$$\Pi = \{\pi_1, \pi_2, \ldots, \pi_N\}$$

Envrionements to Match

$$\mathcal{E} = \{e_1, e_2, \ldots, e_M\}$$

Scoring Function

$$\text{score}(e, \pi) \in \mathbb{R}$$

Batch Scoring at Each Training Step

$$\mathbf{S} = \begin{bmatrix} \text{score}(e_1, \pi_1) & \cdots & \text{score}(e_1, \pi_N) \\ \vdots & \ddots & \vdots \\ \text{score}(e_M, \pi_1) & \cdots & \text{score}(e_M, \pi_N) \end{bmatrix} \in \mathbb{R}^{M \times N}$$

Basis for Assignments

METHODS

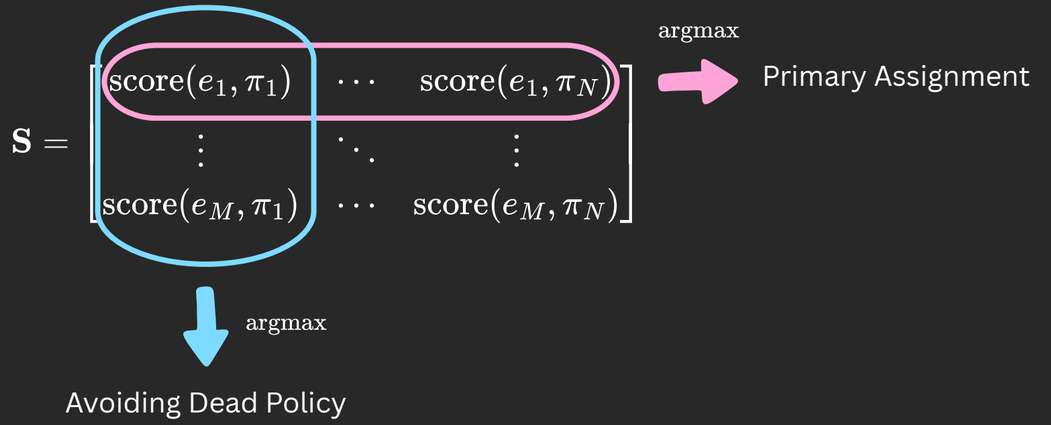
ENVIRONEMNT ↔ POLICY ASSINGMENT

Deriving Assignment Mask Based on Scores

1. Assignment of Each Environment to the Policy It Has Highest Affinity With

2. Assign Policies not Chosen By Any Environments To Pick One They Have Highest Affinity With

3. Final Binary Mask is Obtained by Merging the Two





METHODS

POLICY REPOSITORY: SCORING FUNCTIONS

Instrinsic Goal: **Deriving the policy to specialization**

Basis of Scoring

- **Policy Performance On The Environment**
  Assign to policy with highest performance during evaluation trials
- **Similarity With Previous Policy Encounters**
  Similarity with embedding of environments that policy has solved in the past

METHODS

POLICY REPOSITORY: SCORING FUNCTIONS

Instrinsic Goal: **Deriving the policy to specialization**

Basis of Scoring

- **Policy Performance On The Environment**
  - High Bar
- **Similarity With Previous Policy Encounters**
  - Low-Cost Approach
- **Dense Assignment / Random**
  - Baseline / Ablation



High Sample &

Computation Cost

(if on-policy)

METHODS

SCORING FUNCTIONS: BASED ON EVALUATION PERFORMANCE

Evaluate frozen policy on the environment

(Fixed # of steps)

Score:

Average return [or solve rate]

across completed episodes

- Sample / computation cost scales linearly by # policies and environments if on-policy
- Frozen evaluation might not represent a policy's "potential as a starting point" to continue training on

METHODS

SCORING FUNCTIONS: BASED ON PAST POLICY ENCOUNTERS

History of past trainings for a policy

$$\mathcal{H}_\pi = \{(\mathbf{e}_h, s_h, a_h)\}_{h=1}^{H}$$

Across all previous encounters with envs

- a_h: Age of the Encounter (in training steps)
- s_h: Policy Solve Rate
- $\mathbf{e}_h$: Environment Embedding

Average of Closeness with Previous Environments in Embedding Space Weighted by Policy Success on Them

$$\text{score}(e, \pi) = \frac{\sum_h w_h \cdot K(e, e_h)}{\sum_h w_h} \quad \text{where} \begin{cases} K(e, e_h) = \exp(-\frac{\|e - e_h\|^2}{2\sigma^2}) \\ w_h = s_h \cdot e^{-\gamma a_h} \end{cases}$$

Decay of Past Experiences Weight (by γ)

Using RBF Kernel as Similarity Metric → Adjustable Cutoff Distance (by σ)

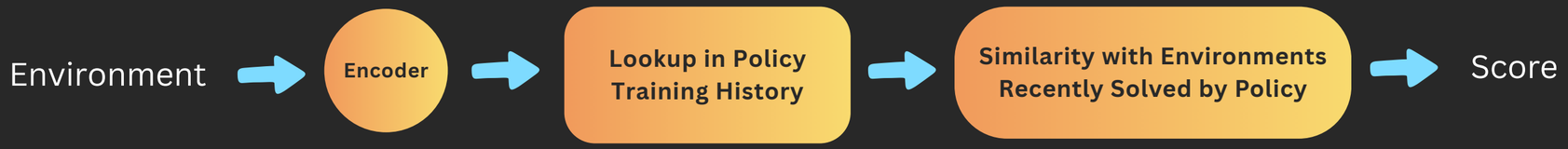
Denominator Normalization → Comparability of Score with Other Policies

METHODS

SCORING FUNCTIONS: BASED ON PAST POLICY ENCOUNTERS

Underlying Assumption:

The more similar an environment's embedding is to those of recently solved environments by the policy, the higher its likelihood of being solved.



How to Encode Environments to Embeddings?

A simplistic approach for fully observable environments such as Kinetix:

Encoding of initial observation state

METHODS

SCORING FUNCTIONS: BASED ON PAST POLICY ENCOUNTERS

1. Object counts in the scene

$$e = \begin{bmatrix} \#\ \text{circles} \\ \#\ \text{polygons} \\ \#\ \text{actuators} \\ \#\ \text{red objects} \\ \dots \end{bmatrix}$$



Exclusive to Kinetix & Maybe its Policy NN Design

But Still Serving as a Simple Surrogate of Observation Space Similarity / Agent Morphology

METHODS

SCORING FUNCTIONS: BASED ON PAST POLICY ENCOUNTERS

1. Object counts in the scene
2. DINOv3 representations of environment render



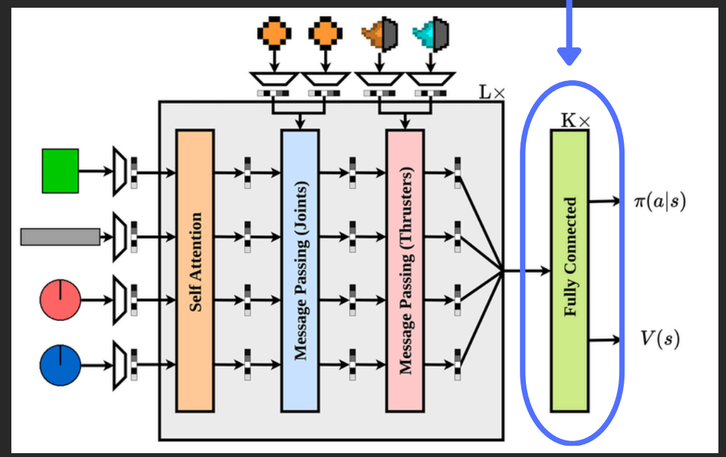
- Assuming visual similarity often leads to policy paramterization similarity
- Using DINO as it preserve visual details rather than high-level semantic information

METHODS

SCORING FUNCTIONS: BASED ON PAST POLICY ENCOUNTERS

1. Object counts in the scene
2. DINOv3 representations of environment render
3. Latent activations of a policy network

Extracted activations
From pretrained policy network
[128 dim] - Normalized L2 norm

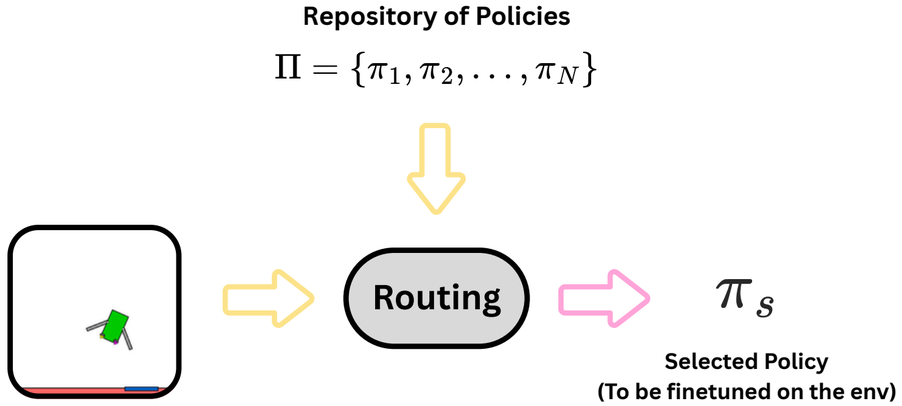


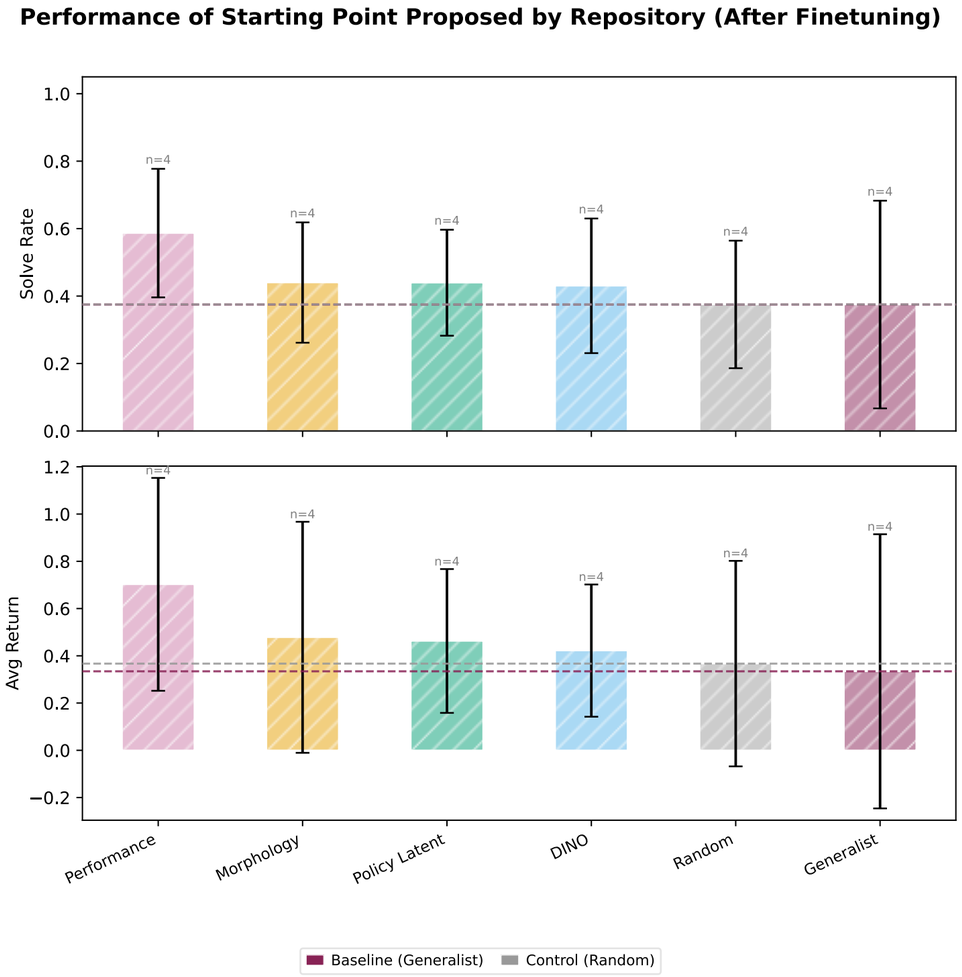
- Obtained from the representation before actor and critic heads
- Ideally containing rich information required to solve the task

RESULTS

EVALUATION ANALYSIS:
PERFORMANCE COMPARISON BETWEEN AFFINITY SCORE CALCULATION LOGICS

- Held-out Set Performance
- Fixed Compute Budget for Finetuning on The Starting Point Proposed by The Scoring Logic





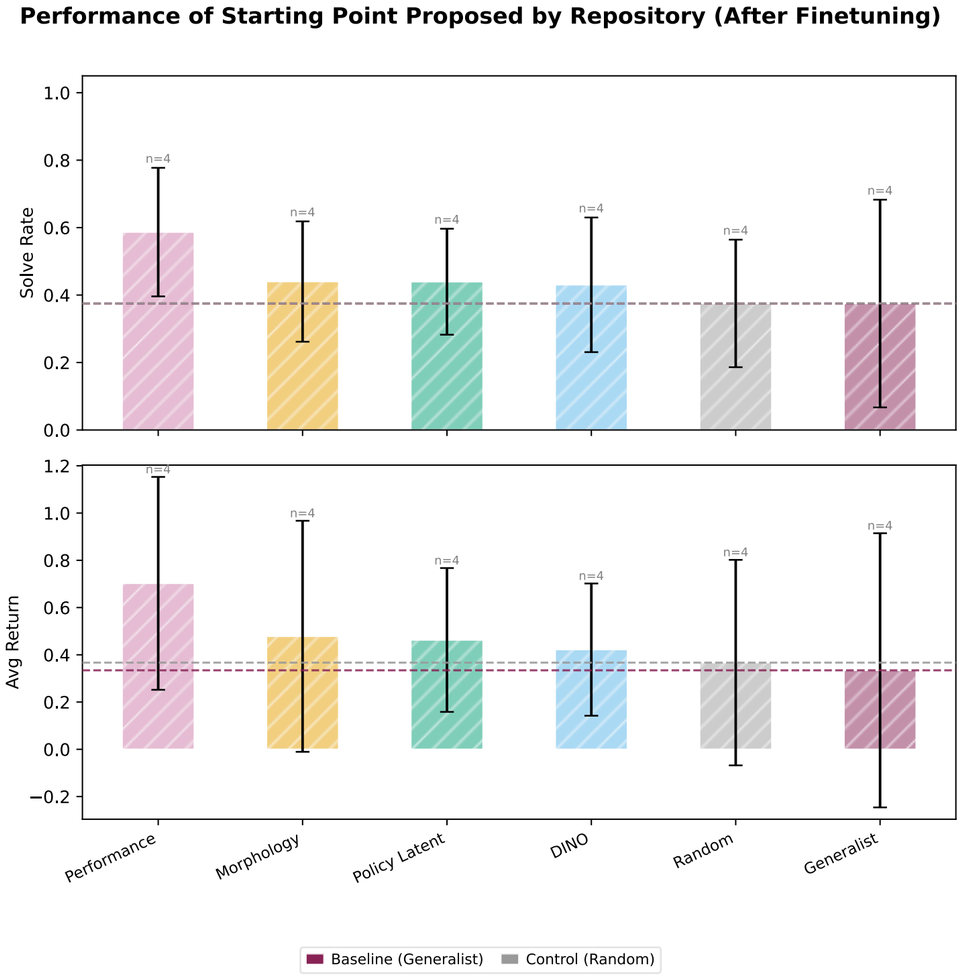
RESULTS

EVALUATION ANALYSIS:
PERFORMANCE COMPARISON BETWEEN AFFINITY SCORE CALCULATION LOGICS

Observations

1. Generalist (trained on all observable environments at each step) and control (random assignment) behave similarly although the higher variance in generalist likely signals the instability caused by continual learning on divergant envrionments.

2. Methods with assignment basis of similarity to environment with the past policy encounters squeeze a bit more performance compared to baseline.



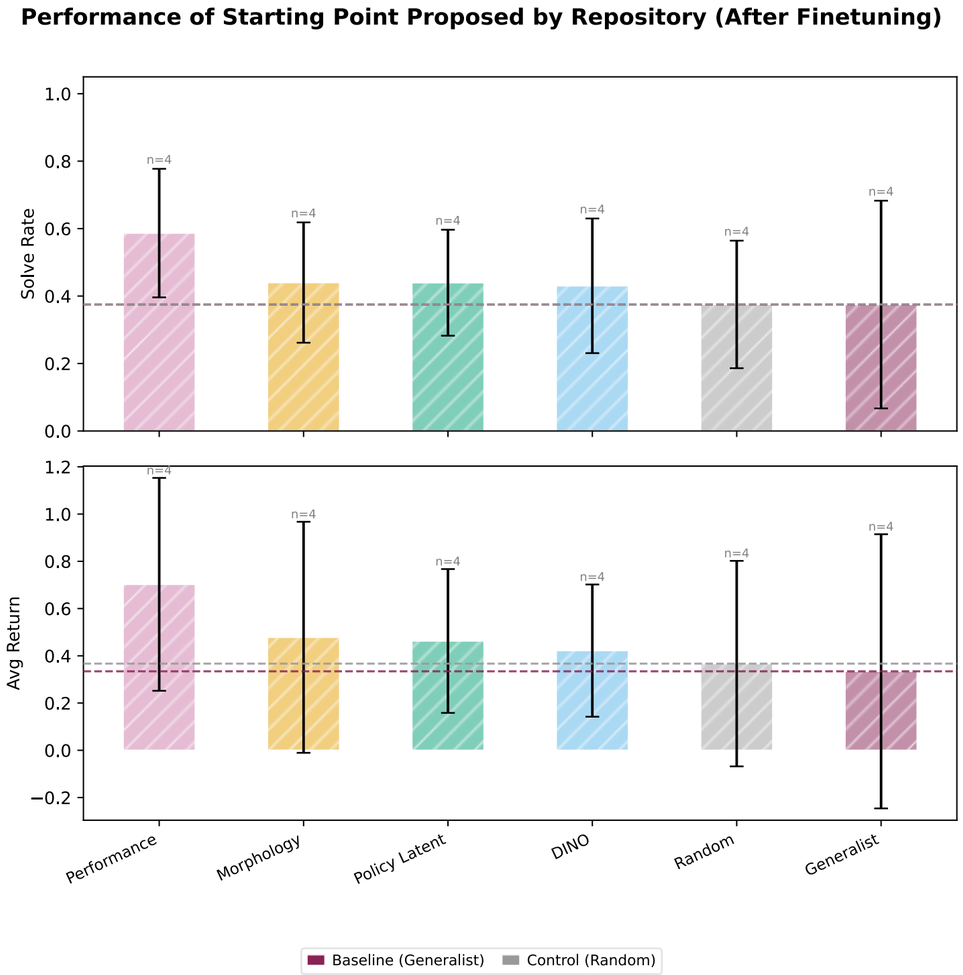
RESULTS

EVALUATION ANALYSIS:
PERFORMANCE COMPARISON BETWEEN AFFINITY SCORE CALCULATION LOGICS

Observations

3. Assigning based on frozen on-policy evaluation return proved to be the best predictor of the performance, although computationally costly

4. Here the visual, policy latent activation and agent morphology show similar performance gains, perhaps due to all being able to capture the similarity between the perturbed envrionments in this specific simplified setup.

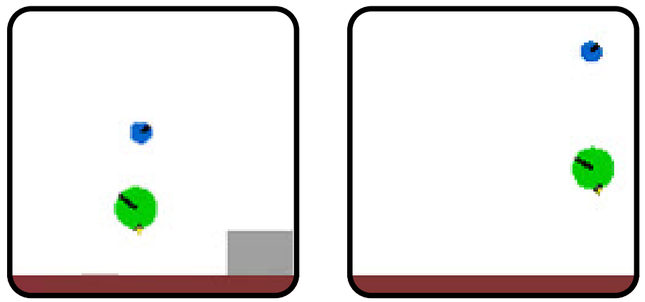


RESULTS

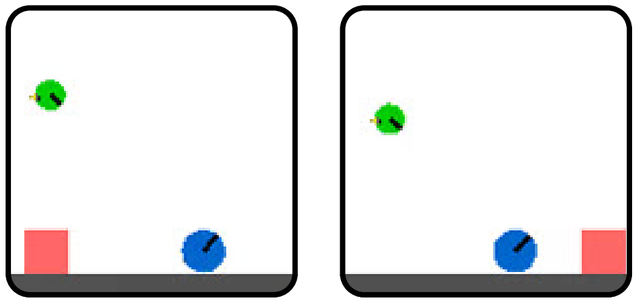
EMERGENT GROUPING OF PERTURBED ENVIRONMENTS

The perturbed environments usually get assigned to the same policy during the evaluation.

But do different scoring logics lead to similar paritioning in the environment space? 🤔



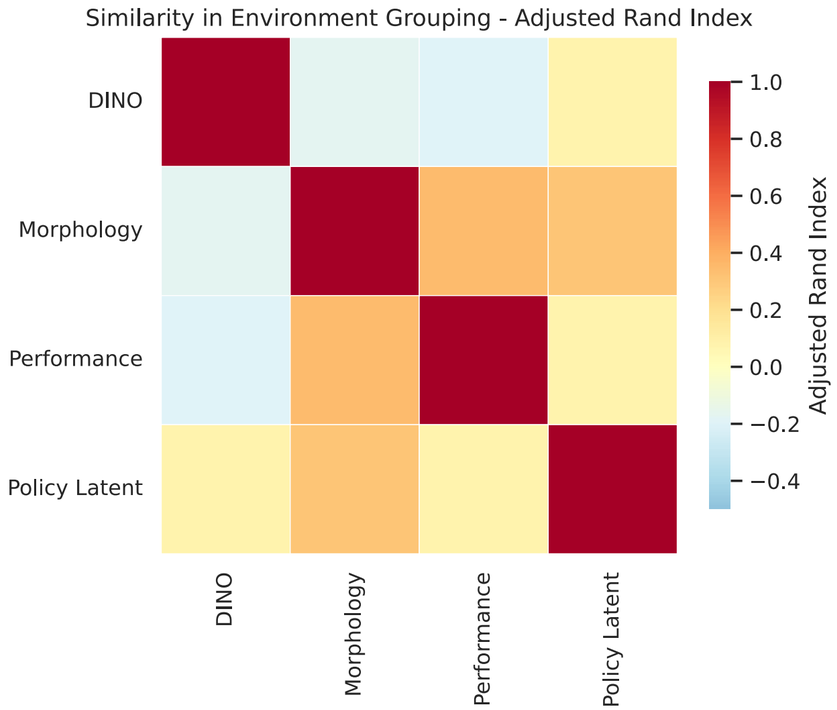
Same policy



Same policy

RESULTS

DIFFERENT SCORING LOGICS LEAD TO DIFFERENT PARTITIONING OF THE ENVIRONMENT SPACE



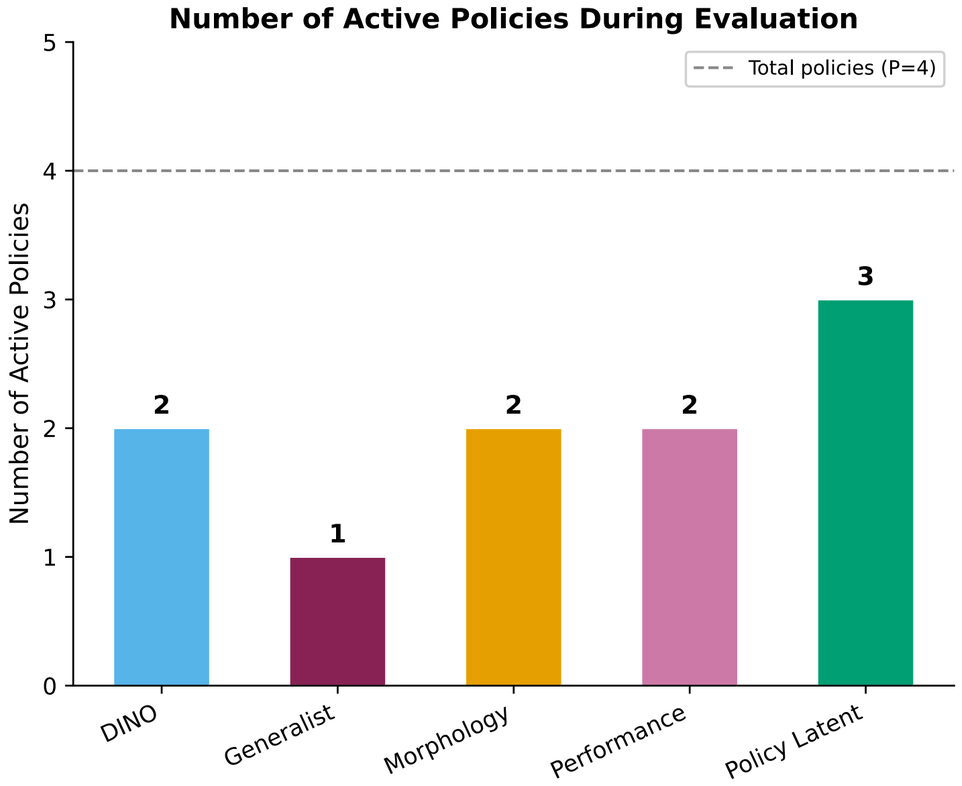
Observations

1. Although the performance of repositories with different assignment logics might be similar, their partitioning of the environment space is necessarily not.

2. Morphology shows above-chance similarity to policy-latent (likely due to policy network treating objects as distinct tokens) and performance based matching.

RESULTS

DOES THE REPOSITORY COLLAPSE TO USING ONE POLICY?



Repositories do not collapse into using a single generalist, serving as a sanity check that the repository framework design leads to partitioning of the environment space.

LIMITATIONS AND FUTURE WORK

- The similarity in environment embedding space might be a transformation of the similarity in policy parameter space

  → Need for learning a mapping between similarity in the two spaces

  → Testing on more diverse perturbations of the environments

- Pruning / Genesis of policies based on usage rate to avoid stagnation over long periods

THANKS!

QUESTIONS & SUGGESIONS?

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